

Inhibition of LPS-Induced TNF- α and NO Production in Mouse Macrophage and Inflammatory Response in Rat Animal Models by a Novel Ayurvedic Formulation, BV-9238

Debendranath Dey, Sunetra Chaskar, Nitin Athavale and Deepa Chitre*

Bioved Pharmaceuticals Pvt. Ltd., BAIF Bhawan, Z Wing, Warje Malwadi, Pune 411 052, India

Rheumatoid arthritis is a chronic crippling disease, where protein-based tumor necrosis factor-alpha (TNF- α) inhibitors show significant relief, but with potentially fatal side effects. A need for a safe, oral, cost-effective small molecule or phyto-pharmaceutical is warranted. BV-9238 is an Ayurvedic poly-herbal formulation containing specialized standardized extracts of *Withania somnifera*, *Boswellia serrata*, *Zingiber officinale* and *Curcuma longa*. The anti-inflammatory and anti-arthritic effects of BV-9238 were evaluated for inhibition of TNF- α and nitric oxide (NO) production, in lipopolysaccharide-stimulated, RAW 264.7, mouse macrophage cell line. BV-9238 reduced TNF- α and NO production, without any cytotoxic effects. Subsequently, the formulation was tested in adjuvant-induced arthritis (AIA) and carrageenan-induced paw edema (CPE) rat animal models. AIA was induced in rats by injecting Freund's complete adjuvant intra-dermally in the paw, and BV-9238 and controls were administered orally for 21 days. Arthritic scores in AIA study and inflamed paw volume in CPE study were significantly reduced upon treatment with BV-9238. These results suggest that the anti-inflammatory and anti-arthritic effects of BV-9238 are due to its inhibition of TNF- α , and NO, and this formulation shows promise as an alternate therapy for inflammatory disorders where TNF- α and NO play important roles. Copyright © 2014 John Wiley & Sons, Ltd.

Keywords: herbal; anti-inflammatory; TNF- α ; NO; adjuvant-induced arthritis rat model; rheumatoid arthritis.

INTRODUCTION

Rheumatoid arthritis (RA) is a chronic immune system-mediated disease characterized by chronic synovitis, which leads to cartilage damage and joint destruction (MacNaul *et al.*, 1990). Cytokines such as tumor necrosis factor-alpha (TNF- α), interleukin-1 β (IL-1 β) and IL-6 play a pivotal role in RA pathogenesis (Roumanes *et al.*, 2011). Overproduction of these pro-inflammatory cytokines increases the severity of RA and results in tissue destruction (MacNaul *et al.*, 1990; Ottonello *et al.*, 2002). TNF- α is a major mediator of apoptosis, inflammation and immunity, and has been implicated in the pathological process of a wide spectrum of other human diseases, including sepsis, diabetes, cancer, osteoporosis, multiple sclerosis, inflammatory bowel disease (IBD) and psoriasis (Ettehadi *et al.*, 1994; Dionne *et al.*, 1997; Mirza *et al.*, 2012). Inflammation is also responsible for expression of enzymes such as inducible nitric oxide synthase (iNOS) and thereby regulates the production of nitric oxide (NO) (Grabowski *et al.*, 1996). NO is a pro-inflammatory mediator, which augments inflammation and aggravates RA (Bingham, 2002; Arias-Salvatierra *et al.*, 2011).

The various treatment options available for RA are disease-modifying anti-rheumatic drugs, anti-inflammatory medications, corticosteroids and the biologic agents (Furst,

2010; Planck *et al.*, 2012). A number of biological agents like etanercept, infliximab, abatacept and adalimumab have been successfully launched in the market and are now the primary treatment modality in TNF- α -mediated diseases such as RA and IBD (Fantini, 2003). However, these agents are associated with serious adverse events such as tuberculosis reactivation and other overwhelming infections, and a predisposition to cause malignancies, especially lymphoma (Bongartz *et al.*, 2006; Bernatsky *et al.*, 2010) and can cause death. Besides, routine injections of these medications lead to other complications, and manufacturing, shelf life and global distribution are challenging. Above all, the cost is prohibitive to the masses of the world (Paul *et al.*, 2006). So an oral, cost-effective, easy to store and administer, small molecule TNF- α inhibitor development has taken a lead position in last several years (Li *et al.*, 2002; Davis and Colangelo, 2013).

Similarly, in osteoarthritis (OA), a type of arthritis affecting weight bearing joints, knees and hips in particular, there is an increasing global awareness regarding the toxicity of various treatment options like nonsteroidal anti-inflammatory drugs (NSAIDs), which are cyclooxygenase-1 inhibitors; and selective cyclooxygenase-2 (COX-2) inhibitors like celecoxib (Hawker *et al.*, 2006). Hence, there is need for more safe, natural and affordable alternatives that have the same effect at the cellular level, but without the huge burden of costs and side effects. So far, limited work has been done in modulating TNF- α signaling or COX-1 and COX-2 inhibition with natural products (Rasool and Sabina, 2007; Nair *et al.*, 2011), and no natural products have been successfully launched using this mechanism to treat the specific diseases (Paul *et al.*, 2006).

* Correspondence to: Deepa Chitre, M.D., Bioved Pharmaceuticals, Inc., 1929 O'Toole Way, San Jose, CA 95131, USA.
E-mail: dchitre@bioved.com

Correction added on 13 May 2014, after first online publication. The authors' first names and surnames had been incorrectly identified, and this has now been fixed.

BV-9238 was developed based on concepts of the ancient science of Ayurveda (Chopra, 2000; Chopra *et al.*, 2001). This poly-herbal formulation comprises specialized and proprietary extracts of four medicinal plants namely Ashwagandha (*Withania somnifera*, Family Solanaceae), Shallaki (*Boswellia serrata*, Family Burseraceae), Ginger (*Zingiber officinale*, Family Zingiberaceae) and Turmeric (*Curcuma longa*, Family Zingiberaceae). Some of the plants in BV-9238 have shown beneficial effect individually in treating autoimmune disease conditions (Ammon *et al.*, 1993; Funk *et al.*, 2006; Zhou *et al.*, 2006; Singh *et al.*, 2007) and in OA (Bliddal *et al.*, 2000). Preclinical and clinical studies on the four plants tested individually show potential for arthritis treatment (Khanna *et al.*, 2007; Gayathri *et al.*, 2007; Ramadan *et al.*, 2011; Park *et al.*, 2007). Also, clinical studies using precursor formulations of the same four herbs, RA-1 and RA-11 (Chopra *et al.*, 2000; Chopra *et al.*, 2004), have been found to be safe and effective in treatment of RA and OA.

BV-9238 is a combination of four well-known Ayurvedic plants/herbs. The doses and ratios of each plant were selected from a large spectrum of plants known for their anti-inflammatory and immuno-modulatory properties. These four were selected due to their synergistic effects in inflammatory and degenerative conditions, in different assays. Selection criteria also included activity in different pathways of inflammation, with an intent that this natural formulation would be effective and safe, working in harmony, by different mechanisms of action. Various permutations, combinations and dose ranging studies were conducted in vitro, in vivo and in Phase II clinical studies to arrive at the current dose and ratios of the four plants. A description of all the development work is beyond the scope of this article. Once the optimal combination and dosage were selected, various pre-clinical and clinical studies were conducted to validate its safety and efficacy. The current work cited here is a demonstration of the mechanism of action studies of this four plant combination.

Literature demonstrates the mechanistic studies of some of the individual plants in BV-9238 (Gayathri *et al.*, 2007; Park *et al.*, 2007). However, our traditional sciences document that, many a time, combination of herbs/plants gives a synergistic action for better efficacy and safety, which may not always be the case with single compounds or single plants. The current work is a modest effort to document mechanism of action of a combination formulation, using modern methods of evaluation.

This study was undertaken to examine the anti-inflammatory and anti-arthritis effect of BV-9238 and to elucidate the underlying mechanism for its activity. The release of cytokine (TNF- α) and pro-inflammatory mediator (NO) from lipopolysaccharide (LPS)-induced mouse monocytic-macrophage cell line was assessed to understand the mechanism of action. Further, two established models of arthritis in rats, namely adjuvant-induced arthritis (AIA) and carrageenan-induced paw edema (CPE), were used to address the anti-inflammatory effect of this novel formulation after oral administration.

MATERIALS AND METHODS

Herbal extract. BV-9238 is a combination of four specialized, cultivated and standardized Ayurvedic plant

extracts, namely *W. somnifera*, *B. serrata*, *Z. officinale* and *C. longa*. The plant materials under study were identified, authenticated and deposited in a Government of India Herbarium, namely Central Council for Research in Ayurvedic Sciences (CCRAS) based in Pune, India. The herbarium voucher numbers are *W. somnifera* (4390), *B. serrata* (4391), *Z. officinale* (4393) and *C. longa* (4392).

Each of the individual plants in BV-9238 and the final formulation was standardized using high pressure liquid chromatography, high pressure thin layer chromatography and potentiometry (Chitre and Dey, 2010). Each of the plant's corresponding standardization markers are provided in parentheses—*W. somnifera* (Withanolide D), *B. serrata* (11-keto Boswellic acid), *C. longa* (Curcuminoids) and *Z. officinale* (Gingerol). The exact percentage of the standard markers is ensured in every single batch of plant extracts for optimal efficacy.

Several separate studies were conducted to evaluate the mechanism of action of the test product. Animal studies were carried out in compliance with Good Laboratory Practices and test guidelines as defined in the European Community guidelines (EEC Directive of 1986; 86/609/EEC). The animal studies were approved and monitored by an in-house Ethics Committee.

Cytotoxicity study. Mouse monocytic-macrophage cell line RAW 264.7 was obtained from American Type Culture Collection (ATCC TIB 71, Rockville, MD, USA). The cells were cultured in 75-cm² plastic flasks (Corning-Costar) with Dubecco's modified Eagle's medium, and supplemented with 10% heat-inactivated fetal calf serum and antibiotics (100 Unitml⁻¹ penicillin and 100 μ g/ml⁻¹ of streptomycin). They were maintained at 37 °C in a humidified atmosphere containing 5% CO₂.

Cell viability was assessed by the mitochondria-dependent reduction of 3-(4,5-dimethyl-thiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) to formazan (Pozzolini *et al.*, 2003). Cells were incubated with different doses (5–2000 μ g/ml) of BV-9238 in 96-well plates at 37 °C for 24 h and with MTT (5 mgml⁻¹) for 4 h. The extent of reduction to formazan within cells was quantified by colorimetric measurement of OD₅₇₀ to determine the cytotoxic dose. BV-9238 was incubated with RAW 264.7 cells for 24 h, in a wide range of doses (0.005–2 mg/ml), and the cytotoxic dose was determined by MTT assay as described above.

LPS-induced TNF- α production. Mouse macrophage RAW 264.7 cells were stimulated with 10 ng/ml LPS (Sigma, St Louis, USA) for 24 h. Cell supernatants were harvested, and TNF-production was detected using Assay Design Enzyme Linked Immunosorbent Assay kit (ELISA) (Assay Designs Inc., Ann Arbor, MI, USA). A seven-point standard curve of two-fold dilutions from 15.63 to 2000 pg/ml of mouse TNF was used. A volume of 100 μ l of the standards and samples of the appropriate dilution was added to a 96-well micro-titer plate. All experiments were performed in triplicate. Further, to determine the kinetics of LPS-induced TNF production, RAW 264.7 cells were cultured with medium, and medium was supplemented with 10 ng/ml LPS in 96-well plates. Cell-free culture supernatants were harvested at 0, 6, 24 and 48 h. The concentrations of TNF- α were evaluated in each culture supernatant using ELISA kit as described above. Further, to determine the effect of the formulation

on LPS-induced TNF- α production, RAW 264.7 cells were stimulated with 10 ng/ml LPS with 500 μ g/ml and 1000 μ g/ml of BV-9238, (two non-toxic doses derived from the cell toxicity study as above), and standard drug dexamethasone (0.3 μ g/ml) for 24 h. At the end of 24 h, media were harvested and TNF levels were measured by ELISA.

LPS-induced NO production. The amount of NO produced was measured by the quantity of nitrite accumulated in the culture medium based on the Griess reaction (Hevel and Marletta, 1997). RAW 264.7 cells were stimulated with LPS (10 ng/ml) for 24 h, 100 μ l of cell culture medium along with BV-9238 1000 μ g/ml and standard drug dexamethasone at 0.3 μ g/ml. Then, the samples of culture media were mixed with 100 μ l of Griess reagent (1% sulphanilamide and 0.1% naphthylethylenediamine dihydrochloride in 2.5% phosphoric acid), the mixture was incubated at room temperature for 10 min and the absorbance at 540 nm was measured in a micro-plate reader (Bio-Rad Laboratories, Hercules, CA, USA). The quantity estimation of nitrite was based on a sodium nitrite standard calibration curve. All experiments were performed in triplicate.

Adjuvant-induced arthritis. Eight-week-old Sprague–Dawley male rats, weighing between 100 and 120 g, were placed in clean polypropylene cages. Animals were maintained on a standard pelleted laboratory animal diet and water *ad libitum* and were housed under standard laboratory conditions at 20 $\pm$ 2 $^{\circ}$ C, 30–70% humidity and 12 h light/dark cycle.

Animals were divided into five groups of six animals each. Arthritis was induced in all rats except Group I by intradermal injection of 0.1 ml of Freund's complete adjuvant [suspension of killed *Mycobacterium tuberculosis* bacteria (0.5% w/w) homogenized in liquid paraffin from Difco Laboratories, USA] in the plantar region of right hind paw of each rat. Group I was sham control, where no adjuvant was injected, Group II was arthritis control (untreated adjuvant injected animals), and Group III and Group IV were treated with reference standards, which received 100 mg/kg body weight orally of aspirin and 50 mg/kg body weight orally of ibuprofen respectively. Group V animals received BV-9238 at a dose of 1 g/kg body weight per oral route.

Drug treatment was started from the initial day, i.e. from the day of Freund's complete adjuvant injection (Day 0), 30 min before adjuvant injection and continued daily till Day 21. Paw volume was measured on Days 4, 8, 14 and 21 using an in-house built plethysmometer, by displaced mercury through a narrow graduated (in ml) glass tube. Body weight was also measured on Days 0, 4, 7, 14 and 21. On Day 22, the animals were anesthetized with diethyl ether, blood was withdrawn through retro-orbital venipuncture of all groups and hematological parameters such as hemoglobin content, total white blood cell (WBC) count, ESR (erythrocyte sedimentation rate) and red blood cell (RBC) count were analyzed.

Carrageenan-induced paw edema in rats. Wistar strain adult male rats weighing 120–150 g were selected randomly and divided into three groups of six each. Group I was the control group, Group II was given standard drug prednisolone at a dose of 1 mg/kg body weight

and Group III received BV-9238 at a dose of 1 g/kg body weight. Rats were fed a standard laboratory diet and housed in polypropylene cages under ideal environmental conditions. All drugs, including standard drug, were administered orally.

Initial paw volumes of both the hind legs of animals were recorded with the help of the in-house built plethysmometer. Carrageenan (Otto Chemie, Pvt. Ltd., Mumbai, India) solution 0.1 ml of 1% w/v in normal saline was injected intradermally into the left foot pad, 1 h after feeding of standard and test drugs to all animals. The right foot served as a self-control to measure the actual swelling. Paw volumes were recorded after 3 h of carrageenan injections.

Statistical analysis. Results were expressed as mean $\pm$ SD. The significance of difference among the groups was assessed using one-way analysis of variance (ANOVA). $P < 0.05$ was considered significant.

RESULTS

Cytotoxicity

Cytotoxicity of BV-9238 was assessed by the MTT assay using RAW 264.7 mouse macrophage cell line. BV-9238, in the dose range of 0.005–2 mg/ml, was found to be non-toxic. At doses above 1 mg/ml, BV-9238 precipitated in tissue culture plates and therefore not used in further studies.

Effect of BV-9238 on TNF- α level

As shown in Fig. 1, LPS stimulation of RAW 264.7 mouse macrophage cells generated a surge of TNF- α production within minutes and continued for at least 48 h. Total TNF- α amount (in pg/ml) was calculated from a standard dilution of known TNF- α as provided by the manufacturer. LPS-treated cells showed a significant increase in TNF- α production compared to untreated cells such that at 24 h there was a seven-fold difference. Subsequent inhibition experiments were carried out at 24-h time point where BV-9238 and dexamethasone were added to cells just before addition of LPS. As shown in Fig. 2, BV-9238 inhibited TNF- α production in a dose-dependent manner. At the higher dose

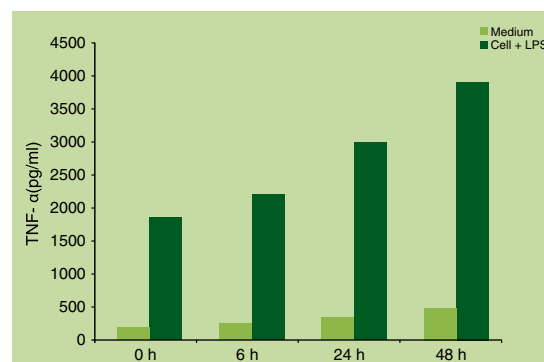


Figure 1. Kinetics of LPS-induced TNF- α production (pg/ml) in RAW 264.7 cells. This figure is available in colour online at wileyonlinelibrary.com/journal/ptr.

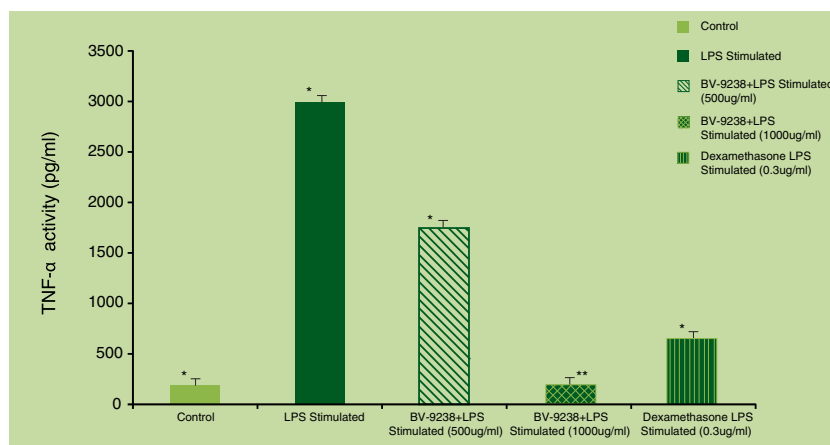


Figure 2. Effect of BV-9238 on TNF- α release by LPS-stimulated macrophages. * $p < 0.05$; ** $p < 0.01$ compared to LPS-stimulation values. This figure is available in colour online at wileyonlinelibrary.com/journal/ptr.

(1000 $\mu\text{g/ml}$), the effect was better than standard drug dexamethasone (0.3 $\mu\text{g/ml}$).

Effect of BV-9238 on LPS-induced NO levels

As LPS induces iNOS, the enzyme responsible for inducible NO production, NO levels in these culture supernatants were measured. As shown in Fig. 3, after 24 h of incubation with LPS there is almost a five-fold increase of NO production compared to untreated control cell supernatants. Pre-incubation with BV-9238 and dexamethasone significantly inhibited the NO production from these cells upon LPS challenge. As described earlier, the effect of dexamethasone in this experiment is not that robust as the dose was kept intentionally very low. BV-9238 at 1000 $\mu\text{g/ml}$ reduced around 50% of NO production in these cells without affecting any cell viability.

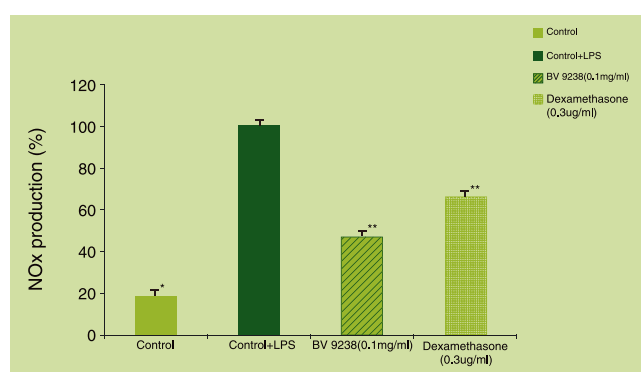


Figure 3. Effect of BV-9238 on NO levels. Data shows mean $\pm$ SD values of triplicate determinations. * $p < 0.05$; ** $p < 0.01$ was calculated by comparing with LPS-stimulation values. This figure is available in colour online at wileyonlinelibrary.com/journal/ptr.

Effect of BV-9238 on adjuvant-induced arthritis in rats

In the AIA model (Winder *et al.*, 1969), with the influence of inflammatory cells, rats develop a chronic swelling in multiple joints, erosion of joint cartilage, bone destruction and remodeling. These inflammatory changes ultimately result in the complete destruction of joint integrity and functions in the affected animal. In this study, either BV-9238 or positive control drugs were administered just before the adjuvant injection. This is considered as the prophylactic model. Ibuprofen has been shown to be effective in this model (Kaur *et al.*, 2011). Aspirin blocks the NF- κB signaling pathway, particularly at the induction step of IKK kinase (Yin *et al.*, 1998), and was administered to one group to see its effect in this prophylactic model as well. The arthritic control group was considered as 100% paw volume. All treatment groups were calculated with reference to arthritic control. BV-9238 inhibited the rat paw edema by 51.2%, which is comparable with standard drugs, aspirin 53.7% and ibuprofen 61.0% after 21 days (Table 1). The control group showed no rat paw edema for 21 days. As shown in Table 2, standard drugs aspirin and ibuprofen, and BV-9238 increased the hemoglobin content when compared to control group. The total WBC counts were remarkably increased in adjuvant-induced rats. However, BV-9238 and the standard drug treated groups normalize the total WBC count when compared to control. The ESR count, which was drastically increased in the arthritic control group, has been remarkably counteracted by the standard drugs aspirin, ibuprofen as well as BV-9238, restoring it back to normal, thus justifying its significant role in inflammatory and arthritic conditions. Further, it is known that these arthritic rats suffer from severe cachexia or body weight loss, but as shown in Fig. 4, there is an improvement of body weight with standard drugs as well as with BV- 9238 treatment.

Table 1. Mean changes in paw volume using plethysmometer in adjuvant-induced arthritis in rats

Treatments	Day 4	Day 8	Day 14	Day 21
Group II (arthritic control)	0.60 $\pm$ 0.02 **	0.90 $\pm$ 0.18 **	1.40 $\pm$ 0.19 **	2.30 $\pm$ 0.17 **
Group III (aspirin)	0.39 $\pm$ 0.04 **	0.48 $\pm$ 0.02 **	0.66 $\pm$ 0.03 **	1.06 $\pm$ 0.08 **
Group IV (ibuprofen)	0.37 $\pm$ 0.03 **	0.46 $\pm$ 0.02 **	0.57 $\pm$ 0.03 **	0.90 $\pm$ 0.03 **
Group V (BV-9238)	0.41 $\pm$ 0.03 **	0.50 $\pm$ 0.03 **	0.57 $\pm$ 0.03 **	1.12 $\pm$ 0.12 **

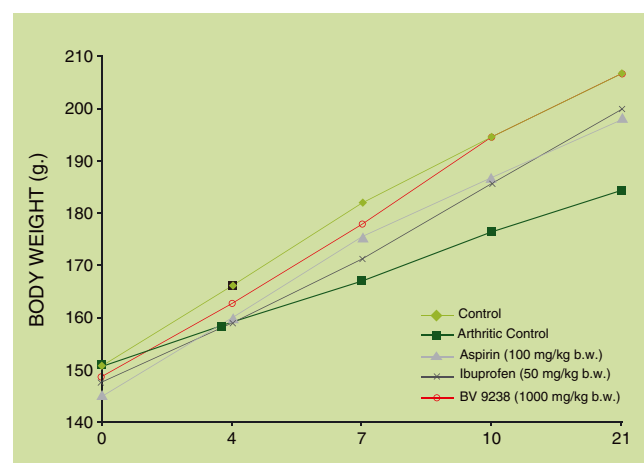
Rats ($n = 6$ in each group). Values are expressed as mean $\pm$ SD. * $p < 0.05$; ** $p < 0.01$.

Table 2. Effect of BV-9238 on hematological parameters in adjuvant-induced arthritis in rats. Rats ($n=6$), values are expressed as mean $\pm$ SD

Treatment	Hemoglobin (Hb gm%)	Total WBC ($\times 10^3$ /c.mm)	RBC count ($\times 10^6$ /c.mm)	ESR (mm/h)
Group I (control)	14.533 $\pm$ 0.58	7.036 $\pm$ 0.48	5.808 $\pm$ 0.28	3.140 $\pm$ 0.17
Group II (arthritis)	13.667 $\pm$ 0.70*	8.129 $\pm$ 0.51*	5.178 $\pm$ 0.06*	5.157 $\pm$ 0.39*
Group III (aspirin)	14.100 $\pm$ 0.64	7.155 $\pm$ 0.60	5.498 $\pm$ 0.28	3.260 $\pm$ 0.10
Group IV (ibuprofen)	14.283 $\pm$ 0.39	7.045 $\pm$ 0.38	5.586 $\pm$ 0.23	3.227 $\pm$ 0.08
Group V (BV-9238)	14.467 $\pm$ 0.45**	7.100 $\pm$ 0.46**	5.702 $\pm$ 0.14**	3.203 $\pm$ 0.10**

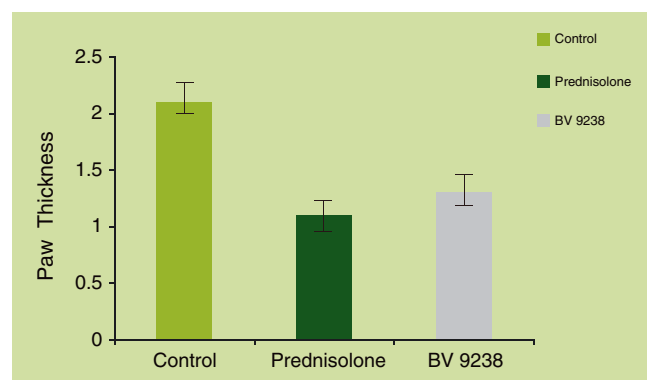
* $p < 0.05$ —significant as compared to Group I (control).

** $p < 0.05$ —significant as compared to Group II (arthritis).

**Figure 4.** Effect of BV-9238 on body weight of adjuvant-induced arthritis in rats. This figure is available in colour online at wileyonlinelibrary.com/journal/ptr.

Effect of BV-9238 on carrageenan-induced paw edema in rats

It is well known for a long time that cyclooxygenase-2 (COX-2) enzyme induction turns on prostaglandin pathway, and CPE in rats characterizes an established model of COX-2-induced inflammation, which shows signs of edema, erythema and hyperalgesia on subcutaneous injection. This model is used to study inflammation in drugs such as NSAIDs and selective COX-2 inhibitors (Guay *et al.*, 2004). Prednisolone (1 mg/kg body weight) and BV-9238 at a dose of 1 g/kg bodyweight was given to rats just before injection of carrageenan. As shown in Fig. 5, in both groups, the inflamed paw thickness was

**Figure 5.** Effect of BV-9238 in carrageenan-induced rat paw edema. * $p < 0.05$ in comparison with Control. This figure is available in colour online at wileyonlinelibrary.com/journal/ptr.

significantly reduced after 3 h of injection, as compared to untreated control animals. Although there is a difference in oral dose between prednisolone and BV-9238, it still clearly shows that this novel herbal formulation may also have some acute inhibitory effect on COX-2-mediated prostaglandin pathway.

DISCUSSION

Traditional Ayurvedic science of medicine has its own method of classifying arthritis into distinct subsets, largely based on the clinical profile of the disease over time and an analysis of certain traits of body and physiology that confer disease susceptibility and pathophysiological responses (Chopra *et al.*, 2001). This complementary and alternative system of medicine can provide effective and safe anti-arthritic drugs, which besides being analgesic and anti-inflammatory can offer chondro-protection. In the present study, we evaluated the anti-inflammatory potential of BV-9238 and the underlying mechanism of action. Initially, we examined the inhibitory effect of the poly-herbal formulation on the production of cytokine TNF- α and pro-inflammatory mediator NO in-vitro and then evaluated its in-vivo anti-inflammatory effects in AIA and CPE rat animal models.

BV-9238 was found to reduce the production of TNF- α and NO levels in the *in-vitro* system using RAW 264.7 cells post LPS stimulation. Dexamethasone, a standard drug, also inhibited the production of TNF- α and NO in LPS stimulated macrophages. Dexamethasone is known to inhibit multiple steps in LPS signaling via the nuclear factor kappa B (NF- κ B) pathway, but the dose used in the study could be a limiting reason for this effect (Mogensen *et al.*, 2008). Results also suggest that BV-9238 inhibited TNF- α production in LPS-stimulated RAW 264.7 cells without causing any cell death. As LPS predominantly activates NF- κ B signalling pathway, which in turn produces TNF- α , present data suggests that BV-9238 may block steps in this signalling pathway. The inhibitory effects of the formulation on NF- κ B pathway may have effects on other pro-inflammatory cytokines like IL-1 β , IL-6, IL-12 and others (Chopra *et al.*, 2000).

In the AIA study, rats were selected because they develop chronic swelling in multiple joints with the influence of inflammatory cells, erosion of joint cartilage and bone destruction (Van Eden and Waksman, 2003; Winder *et al.*, 1969). The model is understood to have close similarities to human rheumatoid disease. The determination of rat paw swelling is a simple, sensitive, quick procedure for evaluating the degree of inflammation and therapeutic effects of drugs. The study showed

that aspirin, ibuprofen and BV-9238 significantly suppressed the swelling in rat paws. Further proof of inhibition of swelling of the rat paw was observed in CPE study using prednisolone (control drug) and test formulation. BV-9238 was found effective in the CPE study, where COX-2-mediated prostaglandin pathways play major role in acute inflammation. Both these inflammatory studies suggest that BV-9238 has a dual action of inhibiting TNF- α and COX-2 activity, as shown by other known small molecules (Pandey *et al.*, 2004).

In arthritic conditions, a mild to moderate rise in WBC count is related to the disease activity and inflammation (Syed and Pinals, 1996). In the present study, as shown in Table 2, aspirin, ibuprofen and BV-9238 show a significant reduction in the total WBC count which indicates suppression of leukocyte migration to the inflamed area. Furthermore, ESR levels increased significantly in the arthritic control group whereas these counts were remarkably controlled in the groups treated with standard drugs aspirin and ibuprofen as well as the group treated with BV-9238. The study highlights the utility of BV-9238 as a potential treatment modality in inflammatory and arthritic conditions.

Decrease in body weight or cachexia has also been used to assess the course of the disease and the response to anti-inflammatory drugs (Winder *et al.*, 1969). As the incidence and severity of arthritis increased, untreated rats underwent significant loss of body weight during the course of the experimental period. However, rats that received standard drugs or BV-9238 showed an increase in body weight during the same experimental period. The increase may be due to restoration of the

absorption capacity of the intestine or due to pain-free general well-being of the animals.

LPS is understood to turn on a number of signaling cascades in NF- κ B pathway and p38 MAP kinase pathways (Wijayanti *et al.*, 2004) but with the absence of further data, it is difficult to postulate the affected step. Further work is under progress to pin-point the effect in IKK Kinase phosphorylation or nuclear translocation of activated p65 subunit. It is worthwhile to mention that in RA patients, an increase in TNF- α levels in synovial fluids of the joints is a critical marker, and it will be worth pursuing further work on the effect of BV-9238 on the endogenous production of TNF- α from synoviocytes (Brennan *et al.*, 1992). Clinical study on a precursor product with the same four natural plant extracts proved safe in long-term administration (Chopra *et al.*, 2000; Chopra *et al.*, 2004). In the current study, the Ayurvedic formulation BV-9238 shows a significant promise for the treatment of disease conditions where TNF- α and COX-2 inhibition play a critical role.

Acknowledgement

The authors would like to thank Dr. Prema R. Rao for her invaluable input in the structure, editing and finalization of the manuscript.

Sponsor—Bioved Pharmaceuticals, Inc.

Conflict of Interest

Dr. Deepa Chitre is a Director of Bioved Pharmaceuticals, Inc. and has shares in the company. Dr. Debendranath Dey has stock options in the company.

REFERENCES

- Ammon HP, Safayhi H, Mack T, Sabieraj J. 1993. Mechanism of anti-inflammatory actions of Curcumin and Boswellic acids. *J Ethnopharmacol* **38**: 113–119.
- Arias-Salvatierra D, Silbergeld EK, Acosta-Saavedra LC, Calderon-Aranda ES. 2011. Role of nitric oxide produced by iNOS through NF- κ B pathway in migration of cerebellar granule neurons induced by lipopolysaccharide. *Cell Signal* **23**: 425–35.
- Bernatsky S, Habel Y, Rahme E. 2010. Observational studies of infections in rheumatoid arthritis: a meta-analysis of tumor necrosis factor antagonists. *J Rheumatol* **37**: 928–931.
- Bingham CO 3rd. 2002. The pathogenesis of rheumatoid arthritis: pivotal cytokines involved in bone degradation and inflammation. *J Rheumatol Suppl* **65**: 3–9.
- Bliddal H, Rosetzky A, Schlichting P, *et al.* 2000. A randomized, placebo-controlled, cross-over study of ginger extracts and ibuprofen in osteoarthritis. *Osteoarthritis Cartilage* **8**: 9–12.
- Bongartz T, Sutton AJ, Sweeting MJ, Buchan I, Matteson EL, Montori V. 2006. Anti-TNF antibody therapy in rheumatoid arthritis and the risk of serious infections and malignancies: systematic review and meta-analysis of rare harmful effects in randomized controlled trials. *JAMA* **295**: 2275–2285.
- Brennan FM, Maini RN, Feldmann M. 1992. TNF- α - a pivotal role in rheumatoid arthritis. *Br J Rheumatol* **31**: 293–298.
- Chitre D, Dey D. Inventors; 2010 Jun 17. Novel formulations to inhibit cyclooxygenase and pro-inflammatory cytokine mediated diseases. US patent application publication 2010/0150865 A1.
- Chopra A. 2000. Ayurvedic medicine and arthritis. *Rheum Dis Clin North Am* **26**: 133–144.
- Chopra A, Lavin P, Patwardhan B, Chitre D. 2000. Randomized double blind trial of an Ayurvedic plant derived formulation for treatment of rheumatoid arthritis. *J Rheumatol* **27**: 1365–1372.
- Chopra A, Patil J, Doiphode V, Patwardhan B. 2001. Exploring ancient Ayurveda for rheumatology; traditional therapy, Modern relevance and challenges. *APLAR J Rheumatol* **4**: 190–199.
- Chopra A, Lavin P, Patwardhan B, Chitre D. 2004. A 32-week randomized, placebo-controlled clinical evaluation of RA-11, an Ayurvedic drug, on osteoarthritis of the knees. *J Clin Rheumatol* **10**: 236–245.
- Davis JM, Colangelo J. 2013. Small-molecule inhibitors of the interaction between TNF and TNFR. *Future Med Chem* **5**: 69–79.
- Dionne S, Hiscott J, D'Agata I, Duhaime A, Sediman EG. 1997. Quantitative PCR analysis of TNF-alpha and IL-1 beta mRNA levels in pediatric IBD mucosal biopsies. *Dig Dis Sci* **42**: 1557–1566.
- Ettahadi P, Greaves MW, Wallach D, Aderka D, Camp RD. 1994. Elevated tumour necrosis factor-alpha (TNF-alpha) biological activity in psoriatic skin lesions. *Clin Exp Immunol* **96**: 146–151.
- Fantini F. 2003. New drugs and treatment strategies for rheumatoid arthritis. *Recent Prog Med* **94**: 361–379.
- Funk JL, Oyarzo JN, Frye JB, *et al.* 2006. Turmeric extracts containing curcuminoids prevent experimental rheumatoid arthritis. *J Nat Prod* **69**: 351–355.
- Furst DE. 2010. Development of TNF inhibitor therapies for the treatment of rheumatoid arthritis. *Clin Exp Rheumatol* **28**: S5–12.
- Gayathri B, Manjula N, Vinaykumar KS, Lakshmi BS, Balakrishnan A. 2007. Pure compound from *Boswellia serrata* extract exhibits anti-inflammatory property in human PBMCs and mouse macrophages through inhibition of TNFalpha, IL-1 beta, NO and MAP kinases. *Int Immunopharmacol* **4**: 473–82.
- Grabowski PS, Macpherson H, Ralston SH. 1996. Nitric oxide production in cells derived from the human joint. *Br J Rheumatol* **35**: 207–212.
- Guay J, Bateman K, Gordon R, Mancini J, Riendeau D. 2004. Carrageenan-induced Paw Edema in Rat Elicits a Predominant Prostaglandin E₂ (PGE₂) Response in the Central Nervous System Associated with the Induction of Microsomal PGE₂ Synthase. *J Biological Chem* **279**: 24866–24872.
- Hawker GA, Jeffrey N Katz JN, Daniel H Solomon DH. 2006. The patient's perspective on the recall of Vioxx. *J of Rheumatol* **33**: 1082–1088.
- Hevel JM, Marletta MA. 1997. Nitric-oxide synthase assay. *Method Enzymol* **233**: 250–258.
- Kaur S, Bijjem K, Sharma P. 2011. Anti-inflammatory and antihyperalgesic effects of the combination of ibuprofen and

- hemin in adjuvant-induced arthritis in the Wistar rat. *Inflammopharmacol* **19**: 265–272.
- Khanna D, Sethi G, Ahn KS, *et al.* 2007. Natural products as a gold mine for arthritis treatment. *Curr Opin Pharmacol* **3**: 344–51.
- Li T, Sen A., Gowri M, *et al.* 2002. Natural product derived small molecule inhibitors for TNF production - SMI compendium on Advances in Anti-Arthritic Agents - SMI's Third Annual Conference, 23–24 April 2002, London, UK.
- MacNaul KL, Hutchinson NI, Parsons JN, Bayne EK, Tocci MJ. 1990. Analysis of IL-1 and TNF alpha gene expression in human rheumatoid synoviocytes and normal monocytes by in situ hybridization. *J Immunol* **145**: 4154–4166.
- Mirza S, Hossain M, Matthews C, Martinez P. 2012. Type 2-diabetes is associated with elevated levels of TNF-alpha, IL-6 and adiponectin and low levels of leptin in a population of Mexican Americans: A cross-sectional study. *Cytokine* **57**: 1,136–142.
- Mogensen TH, Berg RS, Paludan SR, Østergaard L. 2008. Mechanisms of dexamethasone-mediated inhibition of Toll-like receptor signaling induced by *Neisseria meningitidis* and *Streptococcus pneumoniae*. *Infect Immun* **76**: 189–97.
- Nair V, Singh S, Gupta YK. 2011. Evaluation of the disease modifying activity of *Cochicum luteum* Baker in experimental arthritis. *J Ethnopharmacol* **133**(2): 303–307.
- Ottonello L, Cutolo M, Frumento G, *et al.* 2002. Synovial fluid from patients with rheumatoid arthritis inhibits neutrophil apoptosis: role of adenosine and proinflammatory cytokines. *Rheumatology* **41**: 1249–1260.
- Pandey B, Nag A, Neogi P, *et al.* 2004. BLX-1031, an orally active, novel dual inhibitor of COX-2 and TNF- for the treatment of inflammatory disorders. *FOCIS/ Immunology World Congress*: 197–202.
- Paul AT, Gohil VM, Bhutani KK. 2006. Modulating TNF- α signalling with natural products. *Drug Discov Today* **11**: 725–732.
- Park C, Moon DO, Choi IW, *et al.* 2007. Curcumin induces apoptosis and inhibits prostaglandin E(2) production in synovial fibroblasts of patients with rheumatoid arthritis. *Int J Mol Med* **20**(3): 365–72.
- Planck SR, Woods A, Clowers JS, Nicklin MJ, Rosenbaum JT, Rosenzweig HL. 2012. Impact of IL-1 signalling on experimental uveitis and arthritis. *Ann Rheum Dis* **71**: 753–60.
- Pozzolini M, Scarfi S, Benatti U, Giovine M. 2003. Interference in MTT cell viability assay in activated macrophage cell line. *Anal Biochem* **313**: 338–341.
- Ramadan G, Al-Kahtani MA, El-Sayed WM. 2011. Anti-inflammatory and anti-oxidant properties of *Curcuma longa* (turmeric) versus *Zingiber officinale* (ginger) rhizomes in rat adjuvant-induced arthritis. *Inflammation* **34**(4): 291–301.
- Rasool M, Sabina EP. 2007. Anti-inflammatory Effect of the Indian Ayurvedic Herbal Formulation Triphala on Adjuvant-induced Arthritis in Mice. *Phytother Res* **21**: 889–894.
- Roumanes D, Kobie J, Cavenaugh J, Quataert S, Mosmann T. 2011. T cell responses in rheumatoid arthritis patients treated with TNF antagonists. *J Immunol* **186**: 107.7
- Singh D, Aggarwal A, Murya R, Naik S. 2007. *Withania somnifera* inhibits NF-kappaB and AP-1 transcription factors in human peripheral blood and synovial fluid mononuclear cells. *Phytother Res* **21**: 905–913.
- Syed KM, Pinals RS. 1996. Leucocytosis in rheumatoid arthritis. *J Clin Rheumatol* **2**: 197–202.
- Van Eden W, Waksman BH. 2003. Immune regulation in adjuvant induced arthritis: possible implications for innovative therapeutic strategies in arthritis. *Arthritis Rheum* **48**: 1788–96
- Wijayanti N, Huber S, Samoylenko A, Kietzmann T, Immenschuh S. 2004. Role of NF-kappaB and p38 MAP kinase signaling pathways in the lipopolysaccharide-dependent activation of heme oxygenase-1 gene expression. *Antioxid Redox Signal* **6**: 802–10.
- Winder CV, Stephens LA, Stephens MD. 1969. Comparative bioassays of drugs in adjuvant induced arthritis in rats: flufenamic acid, mefenamic acid and phenyl butazone. *Arthritis Rheum* **12**: 472–481.
- Yin MJ, Yamamoto Y, Gaynor RB. 1998. The anti-inflammatory agents aspirin and salicylate inhibit the activity of I (kappa) B kinase-beta. *Nature* **396**: 77–80.
- Zhou HL, Deng YM, Xie QMJ. 2006. The modulatory effects of the volatile oil of ginger on the cellular immune response in vitro and in vivo in mice. *J Ethnopharmacol* **105**: 301–305.